



ITsimplera Institute

Red Team Cybersecurity Internship Program

Week 01

Title: Cybersecurity Fundamentals & Security Concepts

Name	Muhammad Awais
Reg	REDB01-0735
Internship Program	Cybersecurity (Red Team)
Instructor	Talal Rehan
Week	01
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Objectives of the assignment:

- ⇒ Learn CIA Triad + cybersecurity basics
- ⇒ Research real cyberattacks, link to CIA violations
- ⇒ Study cybersecurity tools, purpose + use case
- ⇒ Design secure home network diagram
- ⇒ Understand cybersecurity analyst's daily role

Task 01

Five real-world cyberattacks

Attack	Year	Org	Type	Description	Impact	CIA Violated (how)
WannaCry	2017	National Health Service (UK), FedEx, Renault, Telefónica (global)	Ransomware worm (Server Message Block, SMB, exploit)	Self spreading ransomware, used EternalBlue flaw in SMB protocol, encrypted files, demanded Bitcoin ransom	200,000+ machines hit, 150+ countries. Surgeries canceled. Loss \$4B+	Integrity: files encrypted, couldn't be modified/used properly. Availability: systems and hospital services became inaccessible
Equifax Breach	2017	Equifax Inc. (US)	Data breach (Apache Struts web framework exploit)	Unpatched Apache Struts vulnerability let attackers into customer database	143 to 147 million consumers' Personally Identifiable Information (PII), including Social Security Number (SSN) and Date of Birth (DOB), leaked. \$700M+ settlement	Confidentiality: private data stolen and exposed to unauthorized people
Colonial Pipeline	2021	Colonial Pipeline Co. (US)	Ransomware (DarkSide group,	Stolen VPN login gave network	6 days pipeline down, US East Coast fuel	Availability: fuel delivery service

			compromised Virtual Private Network, VPN, credentials)	access, ransomware deployed, fuel operations shut down	shortage, \$4.4M ransom paid	disrupted for days. Confidentiality: company data also exfiltrated before encryption
Target Data Breach	2013	Target Corp. (US retailer)	Data breach (third party vendor credentials led to Point of Sale, POS, malware)	Heating, Ventilation, and Air Conditioning (HVAC) vendor's stolen credentials gave gateway access, malware placed on POS terminals, card data stolen	40M card records plus 70M customer records leaked. \$18.5M multistate settlement	Confidentiality: sensitive card and personal data accessed and stolen without authorization
Marriott to Starwood Breach	2018	Marriott International (Starwood Database, DB)	Data breach (long term unauthorized network access)	Attacker stayed hidden in Starwood network since 2014, later accessed, copied, and encrypted guest reservation database	Around 383M guest records leaked (passport numbers, payment data). £18.4M Information Commissioner's Office (ICO) fine plus \$52M US settlement	Confidentiality: guest personal and travel data accessed and exfiltrated by unauthorized party

Task 02

CIA Triad Component(s)

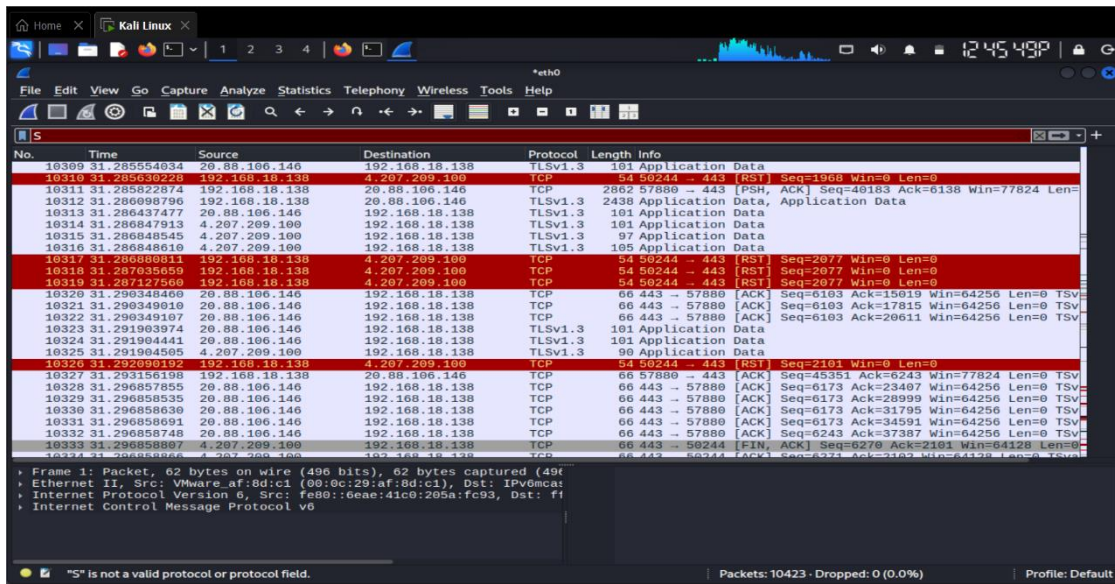
CIA Component	Meaning	Violation Explanation
Confidentiality	Only authorized people can access data	Equifax Breach (2017): 143 to 147 million consumers' Social Security Numbers, birth dates, addresses stolen and exposed to unauthorized parties
Integrity	Data stays accurate and unaltered	WannaCry (2017): ransomware encrypted files across 150+ countries, preventing normal access or modification of that data
Availability	Systems and data stay accessible when needed	Colonial Pipeline Attack (2021): ransomware forced six day pipeline shutdown, blocking legitimate access to fuel delivery services

Task 03

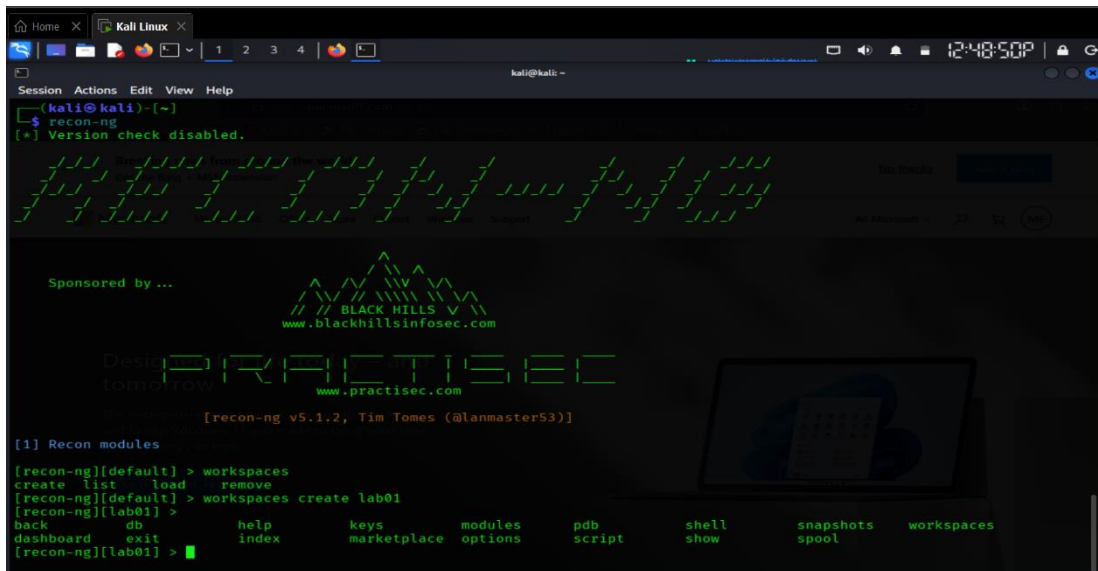
Research 3 cybersecurity tools

Tool	Purpose	Function	Use Case
Wireshark	Network Traffic Analysis and Packet Inspection	Wireshark is an open-source packet analyzer that captures and displays network traffic in real time. It lets analysts inspect individual packets, filter traffic by protocol or IP, and detect unusual patterns, unencrypted data, or suspicious connections	A security analyst uses Wireshark to capture live network traffic and investigate a suspected intrusion, checking for malicious packets or data being sent to an unknown external server
Recon-ng	Open Source Reconnaissance and Information Gathering	Recon-ng is a web reconnaissance framework with a modular, command-line interface similar to Metasploit. It automates gathering of domain info, subdomains, emails, and other public data using various online sources and Application Programming Interfaces (APIs)	A penetration tester uses Recon-ng during the early reconnaissance phase to map out an organization's public-facing digital footprint before planning an attack simulation
Maltego	Link Analysis and Open Source Intelligence (OSINT)	Maltego is a graphical intelligence tool that visually maps relationships between people, domains, IP addresses, companies, and other entities using data pulled from multiple sources	An analyst uses Maltego to visually trace connections between a suspicious domain, its owner, and related infrastructure during a threat investigation

Wireshark:



Recon-ng:

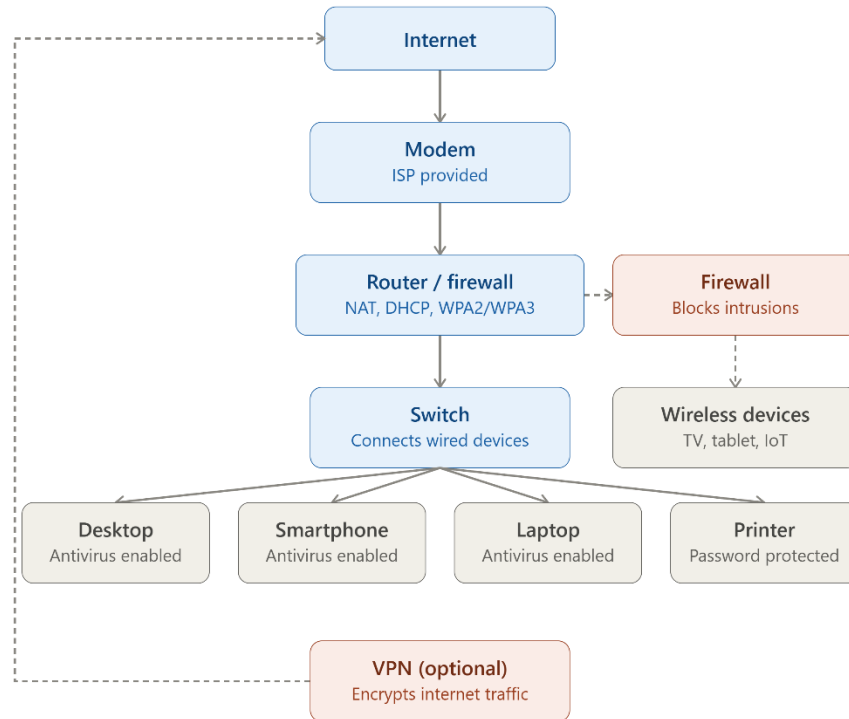


Maltego:



Task 04

Secure home network diagram including Internet, Modem, Router/Firewall, devices, antivirus, firewall, optional VPN.



Task 05

What does a cybersecurity analyst do every day?

A cybersecurity analyst's main job is protecting an organization's computers, networks, and data from cyber threats. Their day usually starts with checking dashboards and alerts to see if anything unusual happened overnight, then moving through a mix of monitoring, investigating, and fixing tasks that keep repeating throughout the day.

Monitoring systems Analysts keep an eye on network traffic and system logs using tools like Security Information and Event Management (SIEM) software, along with Intrusion Detection Systems (IDS) and Endpoint Detection and Response (EDR) platforms. They look for strange login attempts, unusual data transfers, or other warning signs of an attack. Most alerts turn out to be false alarms, but some need deeper investigation, and catching a real threat early can save an organization from major damage later.

Finding weaknesses Analysts regularly scan networks and applications for vulnerabilities before attackers can find them first. Tools like Nmap help discover open ports and exposed services, SQLMap checks web applications for injection flaws, and other scanners look for outdated software or missing security patches. Once weaknesses are found, analysts prioritize them by severity and work with the IT team to apply fixes, patches, or configuration changes.

Responding to incidents When something bad actually happens, like a malware infection, phishing attack, or breach attempt, the analyst jumps into incident response mode. They figure out how the attacker got in, which systems were affected, isolate the compromised devices, remove any malware, and help restore normal operations. Afterward, they write a detailed report explaining what happened, the root cause, and recommendations to prevent it from happening again.

Managing security tools and controls Analysts also configure and maintain firewalls, antivirus software, VPNs, and access control systems. They make sure employees are using strong passwords and multi factor authentication, that data is properly encrypted, and that systems get updated regularly to close security gaps.

Risk assessment and compliance Part of the job also involves evaluating risks and making sure the organization follows security standards and regulations like ISO/IEC 27001, NIST Cybersecurity Framework, or GDPR. This helps the company avoid legal trouble while keeping its systems genuinely secure, not just compliant on paper.

Training and awareness Since technology alone can't stop every attack, analysts also spend time educating other employees. They run short training sessions on spotting phishing emails, creating strong passwords, and following basic security habits, since human error is still one of the biggest causes of breaches.

Staying updated Cyber threats keep evolving, so analysts spend time reading threat intelligence reports, attending workshops, and studying for certifications like Security+, CEH, or CISSP. This keeps them ready for whatever new attack technique comes next.

Summary of week 01 assignment:

This assignment covered cybersecurity fundamentals across five tasks. Real world attacks like WannaCry, Equifax, Colonial Pipeline, Target, and Marriott showed how confidentiality, integrity, and availability get broken through ransomware, stolen credentials, or long term unauthorized access. Mapping each attack to the CIA Triad clarified which security property was actually violated. Tools like Wireshark, Recon-ng, and Maltego showed how analysts monitor, investigate, and gather intelligence. A secure home network design applied the same CIA principles at an individual level using firewalls, antivirus, and an optional VPN. Finally, looking at an analyst's daily routine tied it all together: monitoring, finding weaknesses, responding to incidents, and continuously learning, all built on the same fundamentals.